DSMC Data-Driven Surrogates from Particles to Reproducible Fields

Rarefied cavity, monatomic shock, and diatomic relaxation

Week 10 Extension Lecture Notes

M&I-ENG 690A – AI in Fluid Mechanics

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Central question. How do we qualify noisy particle-solver fields as numerical evidence, then reproduce cavity and shock surrogates without hiding the physical case definition, the assessment metric, or an unavailable target?

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1 Learning Contract and Evidence Map

This lecture accompanies the executable Week 10 notebook and the article by Roohi and Shoja-Sani, *Data-driven surrogate modeling of DSMC solutions using deep neural networks*, Aerospace Science and Technology 168 (2026) 110785, DOI 10.1016/j.ast.2025.110785.

Learning checkpoint

By the end of the lecture, a student should be able to:

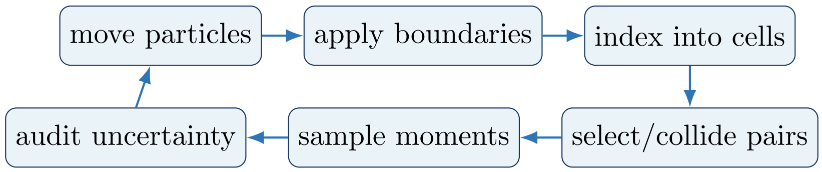
1. summarize the DSMC move–collide–sample cycle;
2. specify mesh, time-step, particle, sampling, and conservation checks;
3. parse a complete numerical field and verify its provenance;
4. derive the log-Kn cavity synthesis used in the article;
5. build and correctly name a POD-polynomial shock-profile surrogate;
6. compare interpolation and one-sided extrapolation; and
7. regenerate every retained metric and figure from committed tables.

Evidence layer	Question	Retained artifact
Raw DSMC	What was actually computed?	85 source files, units, grids, run metadata
Data contract	Did parsing preserve the cases?	SHA-256 manifest and shape checks
Surrogate	What is interpolated, fitted, or extrapolated?	field/profile errors, contours, and an explicit method class
Course release	Can another student rerun it?	notebook, CLI scripts, tests, lecture PDF

2 DSMC as an Independent Numerical Solver

For a dilute gas, $\mathrm{Kn} = \lambda/L$ measures the molecular mean free path λ relative to a characteristic length L. DSMC represents many real molecules by simulation particles and advances them through a statistically equivalent kinetic experiment.

2.1 One DSMC time step



1. **Move:** $\boldsymbol{x}_i^{n+1} = \boldsymbol{x}_i^n + \boldsymbol{c}_i \Delta t$.
2. **Boundary:** impose inlet/outlet models and molecular wall reflection.
3. **Collide:** form local candidates; Bird's NTC rule accepts pairs according to relative speed and cross section.
4. **Internal energy:** for a diatomic gas, a relaxation model transfers energy between translational and rotational modes.
5. **Sample:** after the transient, average number, momentum, energy, stress, and heat-flux moments over many correlated steps.

Evidence gate

Before a DSMC field can become a label, report $\Delta x/\lambda$, $\Delta t/\tau_c$, particles per cell, transient removal, sampling length, independent seeds or block uncertainty, boundary models, and mass/energy balance. A smooth contour is not a resolution study.

3 Keep the Numerical Experiment Separate

The numerical solver and surrogate answer different questions. Their errors must therefore be measured by different experiments.

Object	Required comparison	Typical acceptance evidence
DSMC solver	grid/time/particle/sample refinements and external benchmark	stable observable within reported uncertainty
Parsed dataset	source table versus compact archive	exact case count, shapes, finite values, checksums
Surrogate	untouched complete physical cases	global and local errors, profiles, physics diagnostics

Decision rule

The Week 10 release first validates the tables and numerical contract, then fits the reduced surrogate, and only afterward evaluates the retained cases. The two decisions remain separately inspectable in CSV/JSON files.

3.1 What is actually supplied?

- 14 complete 50×50 cavity fields: lid speeds 10 and 30 m/s;
- $\mathrm{Kn} = 0.001, 0.01, 0.05, 0.1, 0.5, 1, 10$ at each lid speed;
- six 300-point nitrogen-shock profiles, Mach 1.4 through 1.9;
- seven monatomic shock profiles, Mach 1.4 through 2.0;
- wall/heat-flux/run metadata, SPARTA cavity inputs, and a relaxation script.

The paper's 100 m/s cavity and Mach-2 diatomic plots are not numerically scored because those target tables are absent from the supplied archive.

4 Rarefied Cavity: Logarithmic Parameter Synthesis

The development specialists use

$$\mathrm{Kn} \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\},$$

and the complete $\mathrm{Kn} = 0.05$ and 0.5 fields are retained for assessment. For a target Kn_* bracketed by Kn_l and Kn_u,

$$w = \frac{\log_{10} \mathrm{Kn}_* - \log_{10} \mathrm{Kn}_l}{\log_{10} \mathrm{Kn}_u - \log_{10} \mathrm{Kn}_l}, \tag{1}$$

$$\widehat{q}(\mathrm{Kn}_*) = (1 - w)q(\mathrm{Kn}_l) + wq(\mathrm{Kn}_u). \tag{2}$$

This is the article's family-of-specialists synthesis step, applied here directly to complete supplied DSMC spatial fields. Consequently, the sub-2% result measures log-Kn field interpolation; it is not the error of a newly trained spatial specialist. The distinction is part of the learning objective.

For velocity, the normalized RMSE is

$$\mathrm{NRMSE}_U = 100 \frac{\sqrt{N^{-1} \sum_i (\widehat{U}_i - U_i)^2}}{U_{\mathrm{lid}}},$$

while temperature is normalized by the 50 K wall-temperature difference. The denominator is part of the result: a component-relative error can appear large when the component itself has a very small norm.

Scientific boundary

Heat flux and stress are higher moments of the particle distribution and are noisier than velocity and temperature. They remain in the metrics CSV as diagnostics but are not silently averaged into the primary-variable gate.

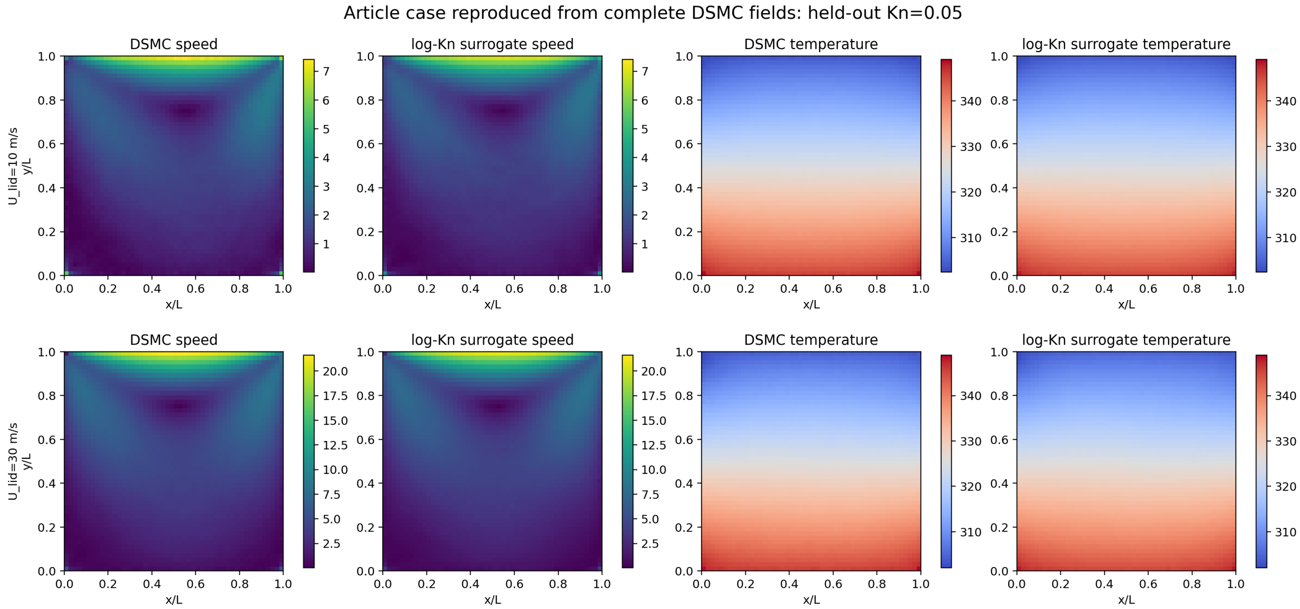


Figure 1: Held-out Kn = 0.05 cavity fields. Reference and prediction use shared color limits; the third panel in each row shows absolute error.

U_{lid}	held-out Kn	U NRMSE	V NRMSE	T NRMSE
10 m/s	0.05	0.895%	1.281%	0.535%
10 m/s	0.50	0.812%	0.784%	0.463%
30 m/s	0.05	0.568%	0.520%	0.584%
30 m/s	0.50	0.622%	0.473%	0.532%

Evidence gate

All twelve primary cavity comparisons pass the predeclared 2% gate. The maximum is 1.281%; the mean is 0.672%. These are errors against the supplied complete DSMC target fields, not values estimated from the published contour.

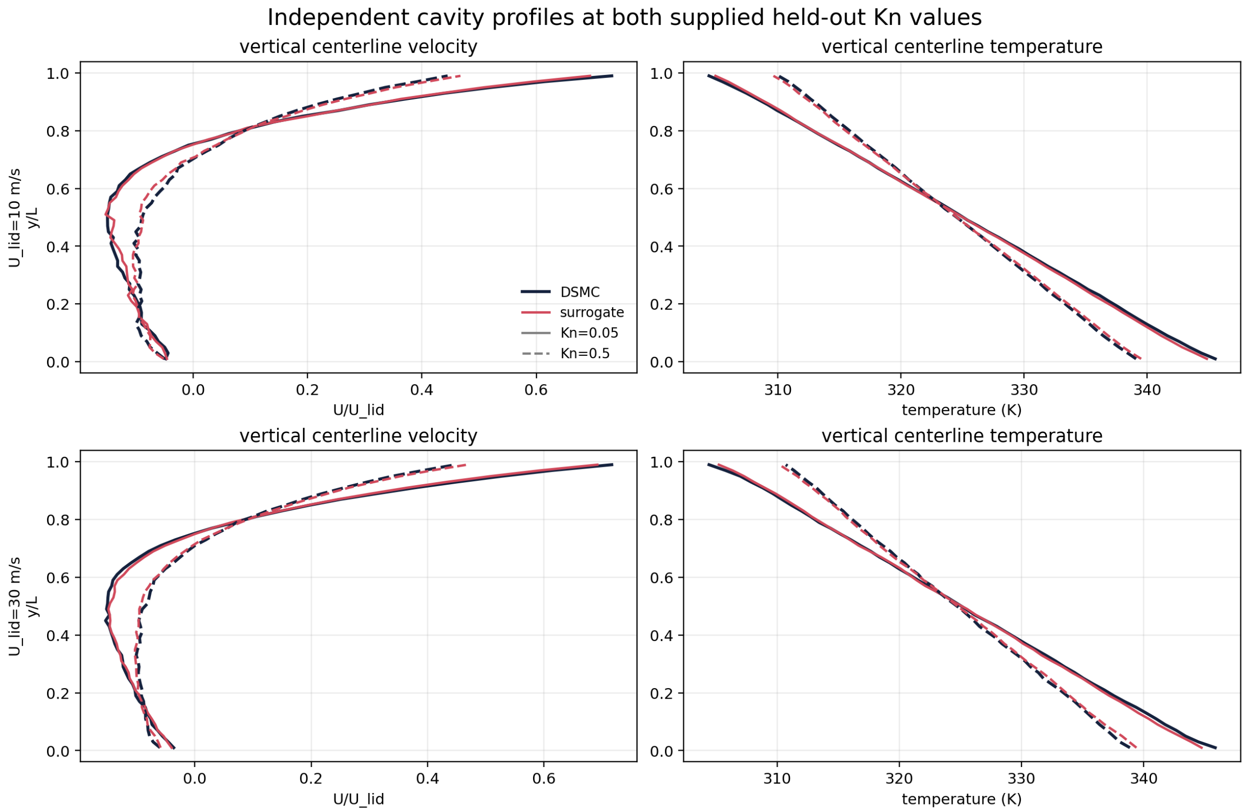


Figure 2: Midline profiles for all four held-out cavity conditions. Profiles expose local bias that a single field norm may hide.

4.1 Student validation checklist

1. Recompute w for both held-out Kn values.
2. Confirm the 50-by-50 ordering against the Tecplot zone declaration.
3. Plot U, V, and T with shared reference/prediction scales.
4. Report RMSE, its normalization scale, and a midline comparison.
5. Inspect q_x, q_y, and τ_{xy} separately; explain their sampling sensitivity instead of merging them into a favorable mean.

5 Diatomic Shock: Translational–Rotational Nonequilibrium

A diatomic shock contains two energy time scales. Translational temperature responds rapidly inside the shock; rotational temperature lags and then relaxes toward equilibrium. The translational overshoot is therefore a central feature, not plotting noise.

The teaching profile surrogate uses POD modes and polynomial Mach coefficients:

$$\widehat{q}(M,x) = \bar{q}(x) + \sum_{k=1}^{r} b_k(M)\,\phi_k(x). \tag{3}$$

The modes $\phi_k(x)$ are learned only from development profiles; a polynomial maps Mach number to modal coefficients. With only five source profiles and a degree-three coefficient fit, this is a compact POD-polynomial interpolator, not a trained neural branch–trunk operator. Mach 1.7 is retained for interpolation. Mach 1.4 is then retained while Mach 1.5–1.9 train a one-sided extrapolation.

Task	Development Mach values	Target
Interpolation	1.4, 1.5, 1.6, 1.8, 1.9	1.7
One-sided extrapolation	1.5, 1.6, 1.7, 1.8, 1.9	1.4

Scientific boundary

The source names `M14.5` through `M19.5` denote Mach 1.4–1.9; the trailing ".5" is an internal DSMC setting. Treating them as Mach 14.5–19.5 would change the physical problem by an order of magnitude.

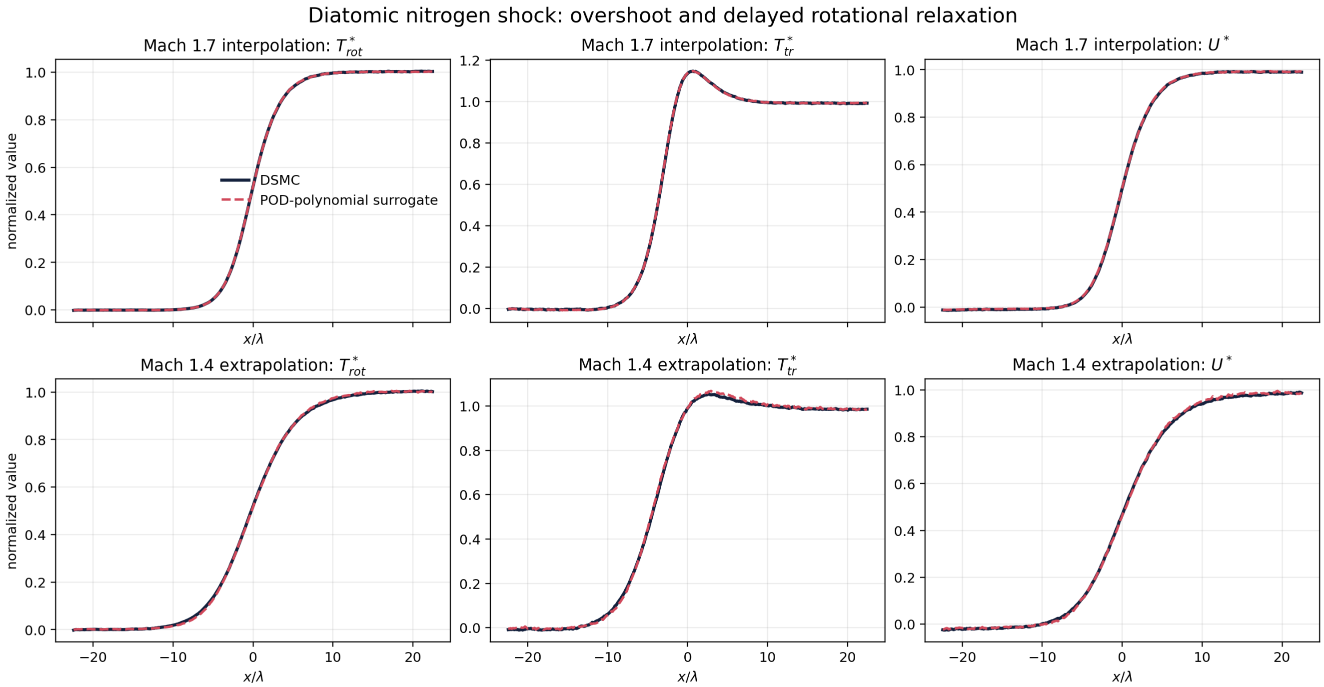


Figure 3: Reproduced diatomic nitrogen-shock profiles. The POD-polynomial surrogate preserves the translational overshoot and delayed rotational relaxation.

Field	Mach-1.7 interpolation	Mach-1.4 extrapolation
Rotational temperature	0.142%	0.630%
Translational temperature	0.260%	1.018%
Normalized velocity	0.291%	0.926%

Evidence gate

All six diatomic profile comparisons pass the 1.5% relative-L_2 gate. The strongest statement is interpolation at Mach 1.7. The extrapolation result is reported separately and is not promoted to a general out-of-range guarantee.

6 Monatomic Shock and Equilibrium Audit

The monatomic archive provides density, normalized velocity, and temperature profiles from Mach 1.4 through 2.0. Retaining Mach 1.7 gives errors of 0.178%, 0.156%, and 0.210%, respectively.

The Maxwell speed distribution supplies a separate equilibrium check:

$$f(c;T) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} c^2 \exp\left(-\frac{mc^2}{2k_B T}\right), \qquad \int_0^\infty f(c;T)\,dc = 1. \tag{4}$$

At the unseen $T = 325$ K classroom target, numerical quadrature gives a unit integral with absolute error 2.35×10^{-14}.

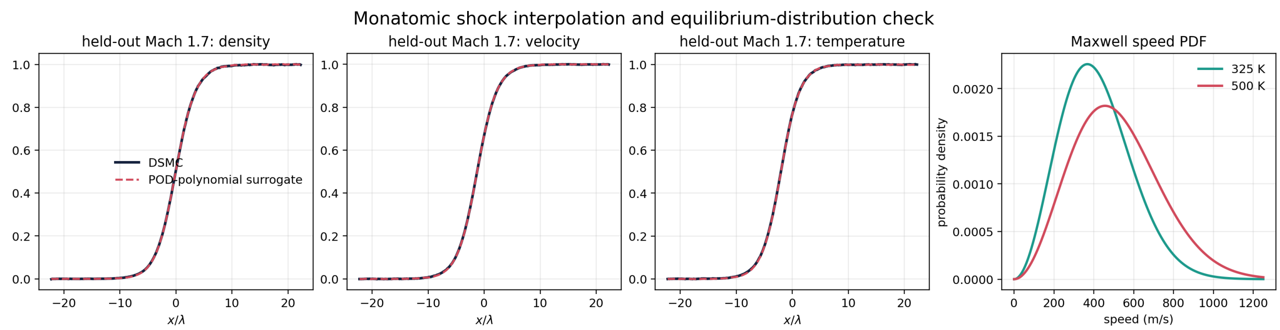


Figure 4: Monatomic Mach-1.7 profile reproduction and the Maxwell speed-PDF normalization check.

7 Combined Release Evidence

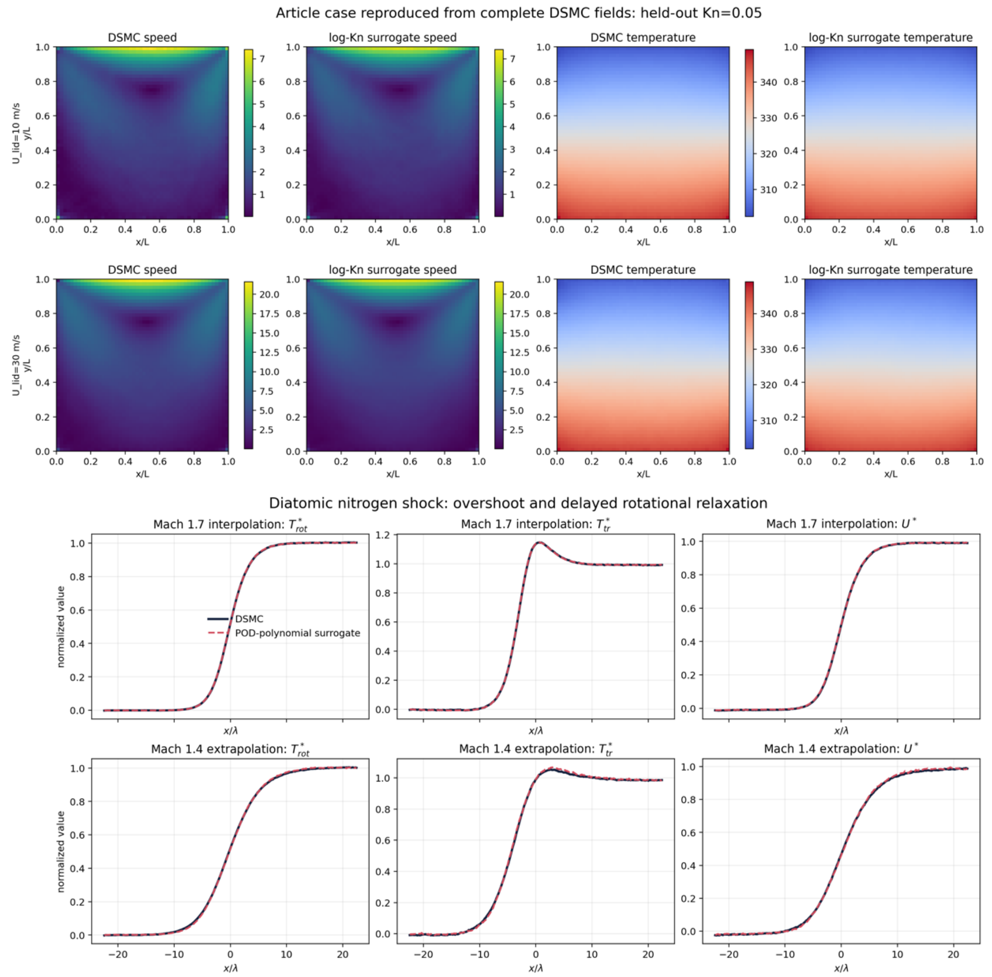


Figure 5: One-page summary regenerated from the committed raw numerical tables.

Release quantity	Retained value
Cavity primary maximum NRMSE	1.281%
Cavity primary mean NRMSE	0.672%
Shock-profile maximum relative L_2	1.018%
Shock-profile mean relative L_2	0.424%
Maxwell PDF normalization absolute error	2.35×10^{-14}

8 Reproduce, Inspect, Extend

From the repository root, the complete evidence chain is:

```
python qa/build_week10_aescte_dsmc_data.py
python qa/run_week10_aescte_validation.py
python -m unittest tests.test_aescte_dsmc -v
```

The first command parses the 85 raw files and writes a checksum manifest. The second regenerates all figures/metrics and enforces the fixed gates. The test suite independently checks shapes, case identities, weights, profile fits, and probability normalization.

Learning checkpoint

Notebook submission.

1. Explain why particle sampling uncertainty precedes surrogate error.
2. Derive the log-Kn weights for 0.05 and 0.5.
3. Report all primary cavity errors and their denominators.
4. Explain the translational-temperature overshoot.
5. Contrast Mach-1.7 interpolation and Mach-1.4 extrapolation.
6. Identify the two article targets absent from the supplied tables.
7. Design one additional DSMC run, including resolution and seed checks.

Decision rule

The final scientific claim is deliberately narrow: the committed tables and scripts reproduce the listed cavity and shock cases within their fixed gates. Any new geometry, lid speed, gas model, or Mach range requires a new numerical qualification and a new case-level assessment.